

Cleaning Data in Pandas

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In [27]: import pandas as pd
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```
In [46]: df = pd.read_excel(r"Customer Call List.xlsx")  
df
```

Out[46]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123/643/9775	93 West Main Street	No	
2	1003	Walter	/White	7066950392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876 678 3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	876 678 3469	98 Clue Drive	N	
8	1009	Gandalf	NaN	N/a	123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	7066950392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876 678 3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123/643/9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	7066950392	3498 Super Lane	Y	
18	1019	Creed	Braton	N/a	N/a	N/a	
19	1020	Anakin	Skywalker	876 678 3469	910 Tatooine Road, Tatooine	Yes	
20	1020	Anakin	Skywalker	876 678 3469	910 Tatooine Road, Tatooine	Yes	

```
In [47]: df = df.drop_duplicates()  
df
```

Out[47]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123/643/9775	93 West Main Street	No	
2	1003	Walter	/White	7066950392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876 678 3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	876 678 3469	98 Clue Drive	N	
8	1009	Gandalf	NaN	N/a	123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	7066950392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876 678 3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123/643/9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	7066950392	3498 Super Lane	Y	
18	1019	Creed	Braton	N/a	N/a	N/a	
19	1020	Anakin	Skywalker	876 678 3469	910 Tatooine Road, Tatooine	Yes	

```
In [48]: df = df.drop(columns = "Not_Useful_Column")
df
```

Out[48]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123/643/9775	93 West Main Street	No	
2	1003	Walter	/White	7066950392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876 678 3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	876 678 3469	98 Clue Drive	N	
8	1009	Gandalf	NaN	N/a	123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	7066950392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876 678 3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123/643/9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	7066950392	3498 Super Lane	Y	
18	1019	Creed	Braton	N/a	N/a	N/a	
19	1020	Anakin	Skywalker	876 678 3469	910 Tatooine Road, Tatooine	Yes	

```
In [49]: #df["Last_Name"] = df["Last_Name"].str.lstrip("...")
#df["Last_Name"] = df["Last_Name"].str.lstrip("/")
#df["Last_Name"] = df["Last_Name"].str.rstrip("_")
#df["Last_Name"] = df["Last_Name"].str.strip("123._/")
df
```

Out[49]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123/643/9775	93 West Main Street	No	
2	1003	Walter	/White	7066950392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876 678 3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	876 678 3469	98 Clue Drive	N	
8	1009	Gandalf	NaN	N/a	123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	7066950392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876 678 3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123/643/9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	7066950392	3498 Super Lane	Y	
18	1019	Creed	Braton	N/a	N/a	N/a	
19	1020	Anakin	Skywalker	876 678 3469	910 Tatooine Road, Tatooine	Yes	

```
In [50]: df["Phone_Number"] = df["Phone_Number"].astype(str).str.strip()

df["Phone_Number"] = df["Phone_Number"].str.replace(r'\D+', '', regex=True)

df["Phone_Number"] = df["Phone_Number"].replace(r'^\s*$', np.nan, regex=True)
df

# ^: The start anchor. It forces the match to look at the very beginning of the str
# \s: Matches any whitespace character (spaces, tabs \t, or newlines \n).
# *: A quantifier meaning "zero or more times". Combined with \s, it means "allow a
# $: The end anchor. It forces the match to look at the very end of the string.
# \D: Matches any character that is not a digit (the uppercase D is the exact oppos
# +: A quantifier meaning "one or more times". It tells regex to grab consecutive n
```

Out[50]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	1235455421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	1236439775	93 West Main Street	No	
2	1003	Walter	/White	7066950392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	1235432345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	8766783469	123 Dragons Road	Y	
5	1006	Ron	Swanson	3047622467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	8766783469	98 Clue Drive	N	
8	1009	Gandalf	NaN	NaN	123 Middle Earth	Yes	
9	1010	Peter	Parker	1235455421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	7066950392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	1235432345	2039 Main Street	Yes	
13	1014	Leslie	Knope	8766783469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	3047622467	214 HR Avenue	N	
15	1016	Ron	Weasley	1235455421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	1236439775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	7066950392	3498 Super Lane	Y	
18	1019	Creed	Braton	NaN	N/a	N/a	
19	1020	Anakin	Skywalker	8766783469	910 Tatooine Road, Tatooine	Yes	

```
In [51]: df['Phone_Number'] = df['Phone_Number'].str.replace(r'(\d{3})(\d{3})(\d{4})', r'\1-  
df                                     # made number like 123-456-789
```

Out[51]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123-643-9775	93 West Main Street	No	
2	1003	Walter	/White	706-695-0392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876-678-3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	876-678-3469	98 Clue Drive	N	
8	1009	Gandalf	NaN	NaN	123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	706-695-0392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876-678-3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123-643-9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	706-695-0392	3498 Super Lane	Y	
18	1019	Creed	Braton	NaN	N/a	N/a	
19	1020	Anakin	Skywalker	876-678-3469	910 Tatooine Road, Tatooine	Yes	

```
In [52]: df[["Street_Address", "State", "Zip_Code"]] = df["Address"].str.split(',', n=2, expand=True)
df
# ',': The delimiter (character) where the split happens.

# 2: The n parameter, which limits the maximum number of splits to 2. This
# ensures the address is cut into a maximum of 3 parts (even if there are more commas)

# expand=True: Forces pandas to return a DataFrame containing separate columns
# instead of a single column containing lists.

# df[["Street_Address", "State", "Zip_Code"]] = ...: Assigns those newly
# generated columns directly back into your original DataFrame df
```

Out[52]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123-643-9775	93 West Main Street	No	
2	1003	Walter	/White	706-695-0392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876-678-3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	876-678-3469	98 Clue Drive	N	
8	1009	Gandalf	NaN	NaN	123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	706-695-0392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876-678-3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123-643-9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	706-695-0392	3498 Super Lane	Y	
18	1019	Creed	Braton	NaN	N/a	N/a	
19	1020	Anakin	Skywalker	876-678-3469	910 Tatooine Road, Tatooine	Yes	

```
In [53]: df["Do_Not_Contact"] = df["Do_Not_Contact"].str.replace('Yes', 'Y')

df["Do_Not_Contact"] = df["Do_Not_Contact"].str.replace('No', 'N')
df
```

Out[53]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123-643-9775	93 West Main Street	No	
2	1003	Walter	/White	706-695-0392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876-678-3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger	NaN	1209 South Street	No	
7	1008	Sherlock	Holmes	876-678-3469	98 Clue Drive	N	
8	1009	Gandalf	NaN	NaN	123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee	NaN	612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	706-695-0392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876-678-3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123-643-9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	706-695-0392	3498 Super Lane	Y	
18	1019	Creed	Braton	NaN	N/a	N/a	
19	1020	Anakin	Skywalker	876-678-3469	910 Tatooine Road, Tatooine	Yes	

```
In [54]: #df = df.replace('N/a', '')
#df = df.replace('NaN', '')

df=df.fillna('') ## This code replaces all missing or null values
df
```

Out[54]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1002	Abed	Nadir	123-643-9775	93 West Main Street	No	
2	1003	Walter	/White	706-695-0392	298 Drugs Driveway	N	
3	1004	Dwight	Schrute	123-543-2345	980 Paper Avenue, Pennsylvania, 18503	Yes	
4	1005	Jon	Snow	876-678-3469	123 Dragons Road	Y	
5	1006	Ron	Swanson	304-762-2467	768 City Parkway	Yes	
6	1007	Jeff	Winger		1209 South Street	No	
7	1008	Sherlock	Holmes	876-678-3469	98 Clue Drive	N	
8	1009	Gandalf			123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee		612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	706-695-0392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876-678-3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123-643-9775	121 Paper Avenue, Pennsylvania	Yes	

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not
17	1018	Clark	Kent	706-695-0392	3498 Super Lane	Y	
18	1019	Creed	Braton		N/a	N/a	
19	1020	Anakin	Skywalker	876-678-3469	910 Tatooine Road, Tatooine	Yes	

```
In [55]: df = df.drop(df[df["Do_Not_Contact"] == 'Y'].index)
df

# df["Do_Not_Contact"] == 'Y': Creates a boolean mask (True/False for every row) ch
# df[df["Do_Not_Contact"] == 'Y']: Filters the DataFrame to isolate only those matc
# .index: Extracts the row index labels of those specific rows.
# df.drop(...): Removes those index labels from the original DataFrame, leaving beh
```

Out[55]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not_
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
2	1003	Walter	/White	706-695-0392	298 Drugs Driveway	N	
4	1005	Jon	Snow	876-678-3469	123 Dragons Road	Y	
6	1007	Jeff	Winger		1209 South Street	No	
7	1008	Sherlock	Holmes	876-678-3469	98 Clue Drive	N	
8	1009	Gandalf			123 Middle Earth	Yes	
9	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
10	1011	Samwise	Gamgee		612 Shire Lane, Shire	Yes	
11	1012	Harry	...Potter	706-695-0392	2394 Hogwarts Avenue	Y	
12	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
13	1014	Leslie	Knope	876-678-3469	343 City Parkway	Yes	
14	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
15	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
16	1017	Michael	Scott	123-643-9775	121 Paper Avenue, Pennsylvania	Yes	
17	1018	Clark	Kent	706-695-0392	3498 Super Lane	Y	
19	1020	Anakin	Skywalker	876-678-3469	910 Tatooine Road, Tatooine	Yes	

```
In [58]: df = df.drop(df[df["Do_Not_Contact"] == ''].index)
# dropping if "Do_Not_Contact" is empty.
df
```

```
Out[58]:
```

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not_
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
2	1005	Jon	Snow	876-678-3469	123 Dragons Road	Y	
3	1007	Jeff	Winger		1209 South Street	No	
4	1008	Sherlock	Holmes	876-678-3469	98 Clue Drive	N	
6	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
7	1011	Samwise	Gamgee		612 Shire Lane, Shire	Yes	
9	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
10	1014	Leslie	Knope	876-678-3469	343 City Parkway	Yes	
11	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
12	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
13	1017	Michael	Scott	123-643-9775	121 Paper Avenue, Pennsylvania	Yes	
15	1020	Anakin	Skywalker	876-678-3469	910 Tatooine Road, Tatooine	Yes	

```
In [60]: df = df.reset_index(drop=True)
#resetting index, unorganized old index, making new sequential index in order
df
```

Out[60]:

	CustomerID	First_Name	Last_Name	Phone_Number	Address	Paying Customer	Do_Not_
0	1001	Frodo	Baggins	123-545-5421	123 Shire Lane, Shire	Yes	
1	1005	Jon	Snow	876-678-3469	123 Dragons Road	Y	
2	1007	Jeff	Winger		1209 South Street	No	
3	1008	Sherlock	Holmes	876-678-3469	98 Clue Drive	N	
4	1010	Peter	Parker	123-545-5421	25th Main Street, New York	Yes	
5	1011	Samwise	Gamgee		612 Shire Lane, Shire	Yes	
6	1013	Don	Draper	123-543-2345	2039 Main Street	Yes	
7	1014	Leslie	Knope	876-678-3469	343 City Parkway	Yes	
8	1015	Toby	Flenderson_	304-762-2467	214 HR Avenue	N	
9	1016	Ron	Weasley	123-545-5421	2395 Hogwarts Avenue	No	
10	1017	Michael	Scott	123-643-9775	121 Paper Avenue, Pennsylvania	Yes	
11	1020	Anakin	Skywalker	876-678-3469	910 Tatooine Road, Tatooine	Yes	



In []: